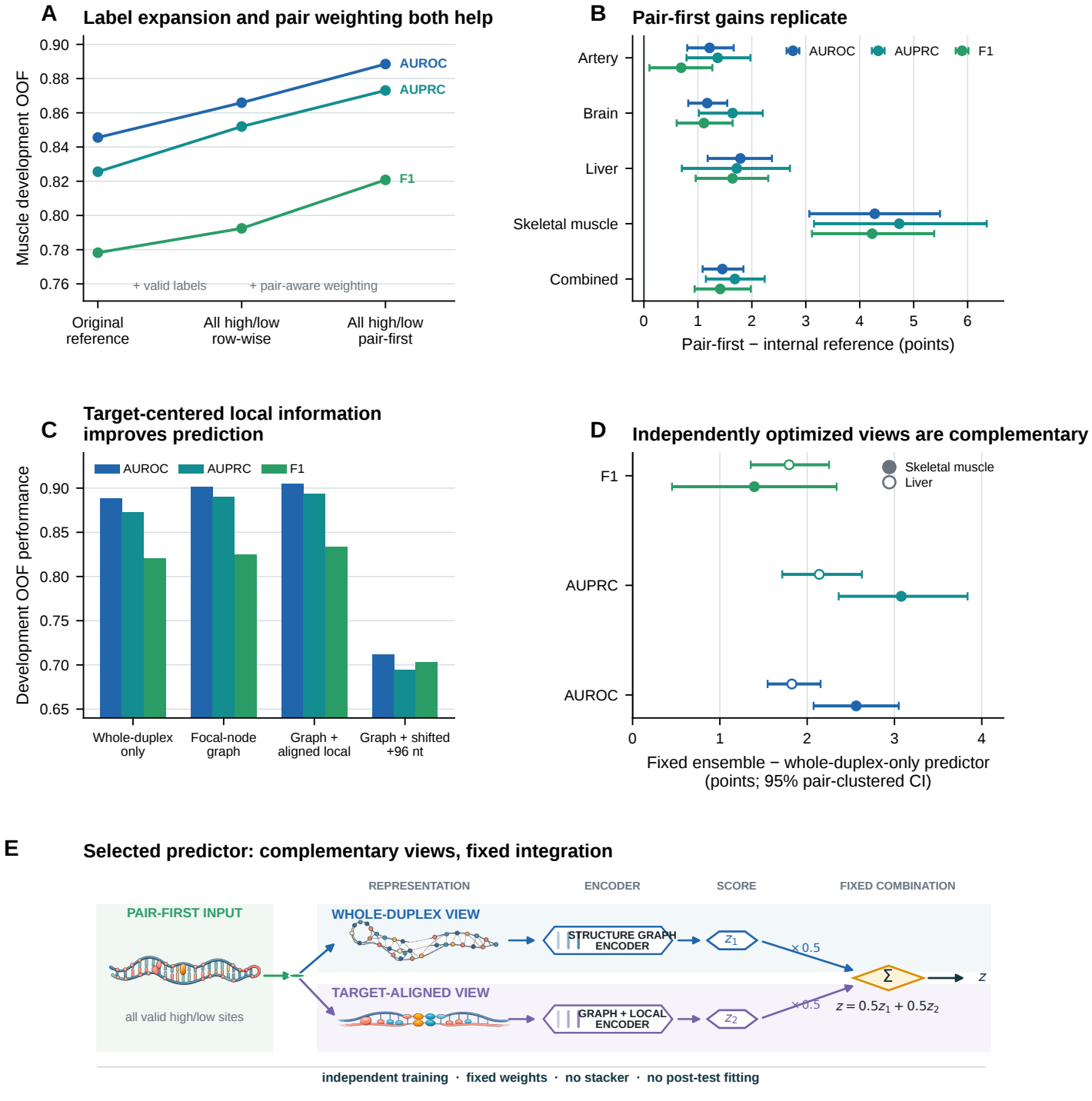

A
Label expansion and pair weighting both help
Muscle development OOF
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
AUROC
AUPRC
F1
+ valid labels
+ pair-aware weighting
Original reference
All high/low row-wise
All high/low pair-first
B
Pair-first gains replicate
AUROC
AUPRC
F1
Artery
Brain
Liver
Skeletal muscle
Combined
0
1
2
3
4
5
6
Pair-first − internal reference (points)
C
Target-centered local information improves prediction
AUROC
AUPRC
F1
Development OOF performance
0.90
0.85
0.80
0.75
0.70
0.65
Whole-duplex only
Focal-node graph
Graph + aligned local
Graph + shifted +96 nt
D
Independently optimized views are complementary
Skeletal muscle
Liver
F1
AUPRC
AUROC
0
1
2
3
4
Fixed ensemble − whole-duplex-only predictor
(points; 95% pair-clustered CI)
E
Selected predictor: complementary views, fixed integration
REPRESENTATION
ENCODER
SCORE
FIXED COMBINATION
PAIR-FIRST INPUT
all valid high/low sites
WHOLE-DUPLEX VIEW
STRUCTURE GRAPH ENCODER
z_1
× 0.5
TARGET-ALIGNED VIEW
GRAPH + LOCAL ENCODER
z_2
× 0.5
Σ
z
$z = 0.5z_1 + 0.5z_2$
independent training · fixed weights · no stacker · no post-test fitting